

Random Signal Theory–EEE 554

Lecture Notes 2

Classes and Moments of Random Variables

[Reading: Kay, Sections 5.5, 5.7, 10.5, 10.7, 6.3-6.8, 11.3-11.9]

Expected Value (mean) of DRVs and CRVs

[Kay, Sections 6.3–6.6, Sections 11.3–11.6]

► Mean: $\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) \, dx$

Center of PMF/PDF around which values of X cluster or weighted average

► For DRV, mean simplifies $\mu_X = \sum_i x_i \Pr(X = x_i)$

⇒ each value weighted by its probability of occurrence

► Mean of a function of a RV: if $Y = g(X) \Rightarrow \mu_Y = \int_{-\infty}^{\infty} g(x) f_X(x) \, dx$

If DRV, $\mu_Y = \sum_i g(x_i) \Pr(X = x_i)$

► Properties of expectation operator

◦ Linearity: $E[aX + b] = aE[X] + b = a\mu_X + b$, if $a, b \in \mathbb{R}$ and deterministic

◦ $E[g(X)] \neq g(E[X])$

► Tabulated means for classes of RVs: [Kay, Table 6.1, Table 11.1]

Variance and Moments

- ▶ $\text{Var}(X) = \sigma_X^2$, is measure of spread of variation of the values of X around its mean

For DRV/CRV, variance is given by

$$\begin{aligned}\sigma_X^2 &= E[(X - \mu_X)^2] = \int_{S_X} (x - \mu_X)^2 f_X(x) \, dx \\ &= E[X^2 - 2\mu_X X + \mu_X^2] = E[X^2] - 2\mu_X E[X] + \mu_X^2 = E[X^2] - \mu_X^2\end{aligned}$$

For DRV, variance simplifies to: $\sigma_X^2 = \sum_i (x_i - \mu_X)^2 \Pr(X = x_i)$

- ▶ Properties of variance: Let $a, b \in \mathbb{R}$ and deterministic
 - $\text{Var}(b) = 0$
 - $\text{Var}(X + b) = \text{Var}(X)$
 - $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Moments and Characteristic Function [Kay, Sections 6.7, 11.7, 11.8]

► n th moment about mean, $E[(X - \mu_X)^n] = \int_{S_X} (x - \mu_X)^n f_X(x) dx$

► n th moment, $E[X^n]$

► Characteristic function:

$$\Phi_X(\omega) = E[e^{j\omega X}] = \int_{S_X} e^{j\omega x} f_X(x) dx$$

⇒ Fourier transform (FT) of PDF

Useful in computing moments since it can be shown that (using FT properties)

$$E[X^n] = \frac{1}{j^n} \frac{d^n}{d\omega^n} \Phi_X(\omega) \Big|_{\omega=0}$$

Chebyshev Inequality:

Use mean and variance to provide a bound on probability

Easier to obtain an estimate of the mean and variance than an estimate of the distribution of random data

$$\Pr(|X - \mu_X| > \gamma) \leq \frac{\sigma_X^2}{\gamma^2}$$

Useful to have a bound on probability based on two parameters that can be estimated

Estimation of Mean and Variance [Kay, Sections 6.8, 11.9]

If M possible values of X , x_i , $i = 1, \dots, M$

Sample Mean

$$\hat{\mu}_X = \frac{1}{M} \sum_{i=1}^M x_i$$

Sample variance

$$\hat{\sigma}_X^2 = \frac{1}{M} \sum_{i=1}^M \left(x_i - \hat{\mu}_X \right)^2 = \frac{1}{M} \sum_{i=1}^M x_i^2 - \hat{\mu}_X^2$$

Monte Carlo Simulations [Kay, Chapter 2]

Use of histogram to obtain PDF of a Gaussian random variable and effect of increasing number of Monte Carlo simulations (realizations)

```
clear; clf; L=130; % # of histogram bins
M=100,000; % # of realizations
mu=5; var=0.1;
% Generate Gaussian RV
x=mu+sqrt(var).*randn(M,1);
hist(x,L) % plot histogram
xlabel('x value'); title(sprintf('%d Monte Carlo simulations', M));
% plot probability of x versus L values of x
[counts,centers]=hist(x,L); plot(centers,counts/M);
title(sprintf('%d Monte Carlo simulations', M));xlabel('x value');ylabel('prob. of x');
```

